Probing the prioritization of multiple spatial locations held in working memory

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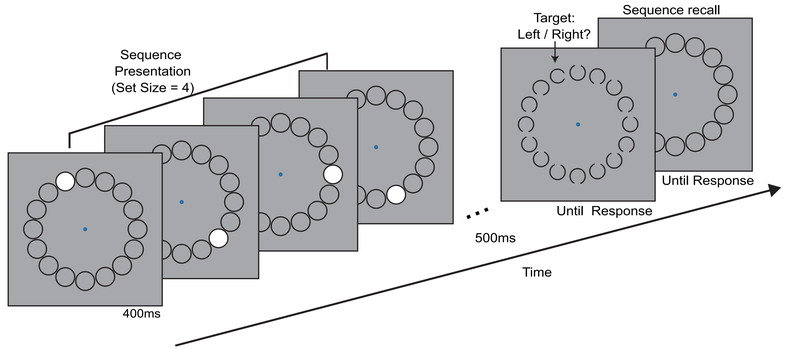
Introduction

- Visual working memory (WM) utilizes attention-based mechanisms to prioritize goal-relevant information (Awh et al., 1998; Oberauer & Hein, 2012; Griffin & Nobre 2003).
- Multiple items prioritized in WM can bias visual search task performance, and facilitate target detection speed (Bahle et al., 2018, 2020; Hollingworth, 2022, Hyunh Cong & Kerzel, 2022).

Current Aim

Determine the degree of attentional prioritization across multiple goal-relevant WM representations.

Methods



Participants

- $N = 24$

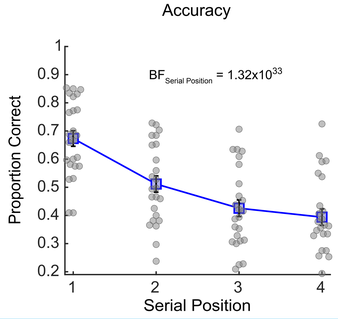
Procedure

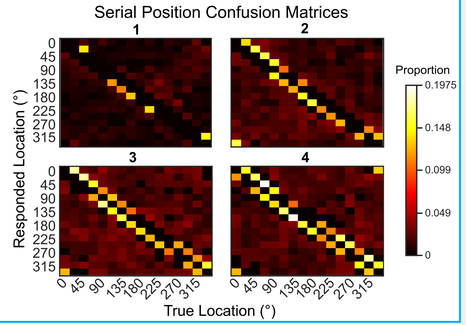
- Covertly memorize sequence of spatial locations.
- Complete visual search task after empty delay period.
- 25% of trials search target appeared in memorized location

Data Acquisition

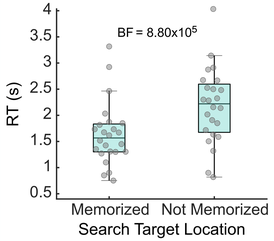
- Eye-tracking (Eyelink 1000) Sampling rate: 500/1000Hz

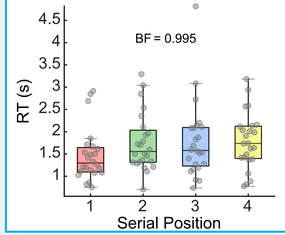
Memory Performance

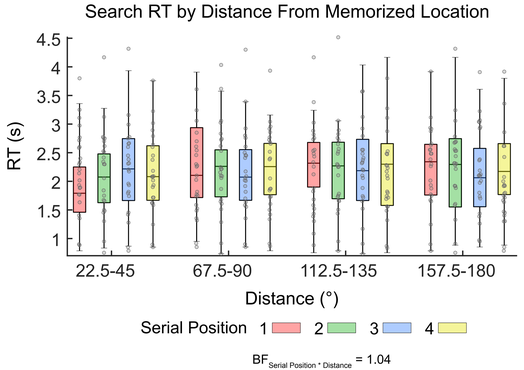




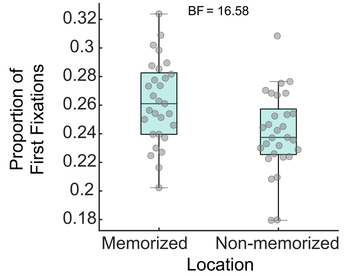
Visual Search Performance

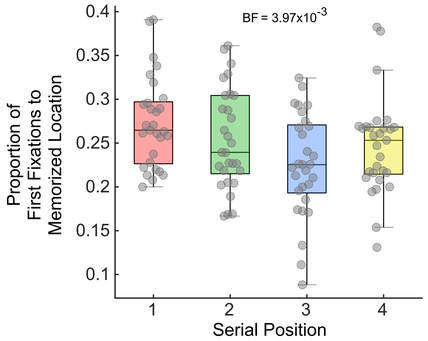






Eye-Tracking





Discussion

- The precision of location representations degrade as serial position increases.
- The degree to which a location is prioritized in WM can influence search speed at other locations that are both near or far away.
- Eye movements are biased toward rememberd locations during visual search.

RT on a secondary task can be used to probe the degree to which an item is prioritized in WM

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